ECE 364: Assignment #3

1. (10 pts) Suppose that in answering a question, in a multiple-choice test, the examinee either knows the answer, with probability p, or she guesses with probability $1 - p$. Assume that the probability of answering a question correctly is 1 for an examinee who knows the answer and $1/m$ for the examinee who guesses, where m is the number of multiple choice alternatives. What is the probability that an examinee knew the answer to a question, given that she has correctly answered it?

2. (10 pts) Suppose we are given a dataset with m descriptive features and a target feature, each with r levels. If the descriptive features are conditionally independent given the level of the target feature, what is the minimum number of probabilities we need to calculate and store from the dataset?

3. (10 pts) Suppose you are given the following training dataset with three Boolean descriptive features, x, y, and z, and Boolean target feature U.

ID	x	y	z	U
1	1	0	0	0
2	0	1	1	0
3	0	0	1	0
4	1	0	0	1
5	0	0	1	1
6	0	1	0	1
7	1	1	0	1

   Suppose you have to predict U using a Naive Bayes model. After training is complete, what would be the predicted probability $P(U = 0|x = 0, y = 1, z = 0)$?

4. (10 pts) The following is a description of the causal relationship between storms, behavior of burglars and dogs, and house alarms: "Stormy nights are rare. Burglary is also rare, and if it is a stormy night, burglars are likely to stay at home. Dogs don't like storms either, and if there is a storm, they like to go inside. Your house alarm is designed to be triggered if a burglar breaks into your house, but sometimes it can be set off by your dog coming into the house, and sometimes it might not be triggered even if a burglar breaks in (the alarm could be faulty or the burglar may be very good)."

   (a) (2 pts) Define the topology of a Bayesian network that encodes these causal relationships.

   (b) (3 pts) The table below lists a set of instances from the house alarm domain. Using the data in this table, create the conditional probability tables (CPTs) for the Bayesian network you created.

   (c) (5 pts) What value will the Bayesian network predict for ALARM, given that there is a storm but we don't know if a burglar has broken in or where the dog is?

ID	STORM	BURGLAR	DOG	ALARM
1	false	false	false	false
2	false	false	false	true
3	false	false	true	false
4	false	true	false	false
5	false	true	false	true
6	false	true	true	true
7	true	false	true	true
8	true	false	true	false
9	true	false	true	false
10	true	true	false	true

5. (20 pts) **Coding project**

   In this project, you will train naive Bayes models and Bayesian networks to determine whether or not a patient has heart disease.

   The dataset consists of descriptive numerical features, including: age in years, trestbps: resting blood pressure, chol: serum cholesterol in mg/dl, thalach: maximum heart rate achieved, oldpeak: ST depression induced by exercise relative to rest, and ca: number of major vessels colored by fluoroscopy.

   The target feature is a binary variable, where 1 indicates the presence of heart disease.

   You will fit naive Bayes models and investigate the effect of different hyperparameters including different feature distributions and strategies for deriving categorical features. You will also fit Bayesian networks and do experiments with networks capturing other feature relationships.

   Please see the Jupyter notebook for more details.

   GitHub repository for ECE364 coding projects: https://github.com/jha-lab/ECE364_2026